

# Scientific Writing Checklist

Page 1/2

Mastering the Scientific Narrative — Skill Session · C. G. Hounmenou (Univ. of Labé / CERFIG)

Tick every box before you submit. Reviewers will check them anyway.

## 1 FRAME — Title, Abstract & Introduction

*The 20% of the paper that 80% of readers will ever see.*

- ☐ **Title names your method, problem** and context (the 15-word promise test).
- ☐ **Abstract makes the 5 moves:** context (stakes + numbers) → gap → contribution → results → impact.
- ☐ **No adjectives without evidence:** every claim of quality is backed by a NUMBER (no “very good / promising”).
- ☐ **Introduction narrows:** general problem → specific gap → your answer (the funnel).
- ☐ **Contributions are** explicitly listed (“Our contributions are threefold: ...”) and each one is verifiable in Results.
- ☐ **Heilmeier questions** answered: what? how is it done today? what is new? who cares — locally AND globally?

## 2 CORE — Method, Data & Reproducibility

*Could a competent stranger rebuild your system from the text alone?*

- ☐ **Every symbol** defined at first use; one notation convention throughout (x, x, X).
- ☐ **Method fully specified:** architecture, losses, optimizer, schedule, hardware, training time.
- ☐ **All hyperparameters** in a table, with the search space stated.
- ☐ **Dataset biography complete:** provenance, size, composition, consent/ethics, preprocessing.
  - ☐ **Splits are separated** by farm/speaker (no leakage!); known biases declared; license stated.
- ☐ **Dataset card / datasheet** published or planned (template in this kit).
- ☐ **Data are F-A-I-R:** DOI (Zenodo), access protocol, standard formats, explicit license.
- ☐ **Code repository ready**, with pinned versions, seeds, configs, and a one-command run.sh.
- ☐ **Repo is anonymized** for review; public + archived (DOI) at camera-ready.

## 3 IMPACT — Results, Figures & Tables

*One figure = one message. Same numbers, different credibility.*

- ☐ **Strong baselines** chosen fairly and compared under identical protocols.
- ☐ **Variance is** reported as mean ± std over ≥ 3 seeds (box plots welcome).
- ☐ **Every figure has** labeled axes with units; full (non-truncated) scales; readable at print size.
- ☐ **Captions are** self-contained: what, where, and the takeaway in the caption itself.
- ☐ **Vector graphics** (PDF/SVG from matplotlib/TikZ) — no blurry screenshots; colorblind-safe palettes.
- ☐ **At least one figure** shows where the model FAILS (confusion matrix, error cases).

### ★ THE FIGURE-1 TEST

Show only your Figure 1 to a colleague for 30 seconds. Can they state your contribution?

If not, redesign Figure 1 before touching anything else — experienced readers look there first.

## 4 HONESTY — Discussion, Limitations & Conclusion

*The trust section. Reviewers respect self-criticism; they punish concealment.*

- ☐ **Discussion** interprets (WHY it works, WHEN it fails) — it does not repeat Results.
- ☐ **Limitations** are specific and honest (e.g., untested conditions, coverage limits, compute constraints).
- ☐ **African-context constraints** framed as observations and future work — not hidden.
- ☐ **Conclusion** restates contribution + strongest number + natural future work.
- ☐ **Broader-impact statement** included where required (dual use, fairness, environmental cost).

## 5 POLISH — Language & Compliance

*Desk rejections are usually mechanical. Eliminate the mechanics.*

- ☐ **Short sentences:** one idea per sentence; active voice preferred; no paragraph longer than ~8 lines.
- ☐ **Overclaiming words** “solve”, “first ever”, “novel” (×5), “obviously” — removed or justified.
- ☐ **Spelling variant** consistent (US or UK) throughout; acronyms defined at first use.
- ☐ **References** are complete, consistent style, no “[?]”; every citation is actually discussed.
- ☐ **Venue rules obeyed:** page limit, template untouched, anonymization respected (no names, no “our previous work [X]”).
- ☐ **Submission** checklist (NeurIPS/ICLR reproducibility & ethics forms) filled honestly.

## 6 BEFORE YOU CLICK “SUBMIT”

*The last 48 hours.*

- ☐ **Paper read by 2 people** who did NOT write it (ideally one expert + one outsider).
- ☐ **You read the abstract** aloud — every stumble is a rewrite signal.
- ☐ **You re-read your own paper** in Pass-1 mode (10 min): does the story survive on title, figures, headings alone?
- ☐ **All figures** checked on the printed page (zoom 100%).
- ☐ **Reproducibility:** run.sh executed on a clean machine by a co-author.
- ☐ **You archived a backup** of the exact submitted version (PDF + source + repo tag).

### 🔪 THE 3-SENTENCE ABSTRACT TEMPLATE (memorize it)

S1 — CONTEXT: “In [region], [problem] causes [stakes + numbers], but existing [solutions] require [barrier].”

S2 — METHOD: “We introduce [Name], a [key property] [system/model] designed for [constraint].”

S3 — IMPACT: “On [data], [Name] achieves [metric = number], improving [baseline] by [number], enabling [impact].”

### → YOUR 6-MONTH CHALLENGE

Week 1–4: 3 papers/week in Pass 1, 1 in Pass 2 — build your literature map (Zotero).

Month 2–3: draft Introduction + Method with the LaTeX template in this kit.

Month 4–5: experiments, figures, internal review. Month 6: SUBMIT (a workshop counts!).